# Hands-On: Types of Router Components and Nested Routing in React

In this lab, you will create a simple React app that uses different types of router components and implements nested routing.

### Step 1: Create a new React app

Create a new React app by running the following command in your terminal:

```
npx create-react-app my-app
cd my-app
```

### Step 2: Install React Router

Install React Router by running the following command in your terminal:

```
npm install react-router-dom
```

### Step 3: Create the components

Create three new components called **`Home`**, **`Products`**, and **`ProductDetails`**. Add the following code to the **`Home.js`** file:

```
import React from 'react';

function Home() {
  return (
    <div>
      <h1>Home</h1>
      <p>Welcome to the home page!</p>
    </div>
  );
}

export default Home;
```

Add the following code to the **`Products.js`** file:

```
import React from 'react';
import { Link, Switch, Route } from 'react-router-dom';
import ProductDetails from './ProductDetails';

function Products({ match }) {
  return (
    <div>
      <h1>Products</h1>
      <ul>
        <li>
          <Link to={`${match.url}/1`}>Product 1</Link>
        </li>
        <li>
          <Link to={`${match.url}/2`}>Product 2</Link>
        </li>
        <li>
          <Link to={`${match.url}/3`}>Product 3</Link>
```

```
      </li>
    </ul>

    <Switch>
      <Route path={`${match.path}/:productId`} component={ProductDetails} />
      <Route path={match.path}>
        <p>Please select a product.</p>
      </Route>
    </Switch>
  </div>
 );
}

export default Products;
```

Add the following code to the `**ProductDetails.js**` file:

```
import React from 'react';

function ProductDetails({ match }) {
 const productId = match.params.productId;

 return (
  <div>
    <h2>Product Details</h2>
    <p>Product ID: {productId}</p>
  </div>
 );
}

export default ProductDetails;
```

**Step 4: Create the routes**

Create a new file called `**App.js**` in the `**src**` directory and add the following code:

```
import React from 'react';
import { BrowserRouter as Router, Route, Link } from 'react-router-dom';
import Home from './Home';
import Products from './Products';

function App() {
 return (
  <Router>
    <div>
      <nav>
        <ul>
          <li>
            <Link to="/">Home</Link>
          </li>
          <li>
            <Link to="/products">Products</Link>
          </li>
```

```
      </ul>
    </nav>

    <Route exact path="/" component={Home} />
    <Route path="/products" component={Products} />
   </div>
  </Router>
 );
}

export default App;
```

This component uses the `**BrowserRouter**` component from React Router to define the routes for the app. The `**Route**` components define the paths and the components to be rendered for each path. The `**Link**` components are used to navigate between the different pages. The **Switch** component is used to render only one of the routes at a time. The `**match**` object is used to get the dynamic parameter from the URL.

**Step 5: Test the app**

Start the app by running the following command in your terminal:

```
npm start
```

Open your browser and navigate to [http://localhost:3000](http://localhost:3000). You should see the home page with a link to the products page. Click on the products link and you should see a list of three products. Click on one of the products and you should see the product details page with the ID of the product.

**Conclusion**

In this lab, you learned how to create a simple React app that uses different types of router components and implements nested routing. You also learned how to define routes, navigate between pages, and pass dynamic parameters using React Router.